

Short Term Undergraduate Software Engineering Training:

Course Design and Practice in China

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Outline.

- Problem and Objectives
- Course Design and Practice
  - Teaching of prerequisite techniques
  - Project based training with agile approach
  - Defects finding between groups
  - Discussion and demonstration
- Evaluation and Result



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Problem and Objectives.

1. Huge gap between what undergraduate students learn and what industry does.
2. Limited practice hours for software engineering courses in China.

Programming	Able to learn a new language or framework in a day. Good coding style.
Engineering Sense	Under stand engineering problems. Requirement analysis? Testing and documentation? Version control?
Individual Abilities	Individual abilities: Requirement Analysis / Software Design / Testing / Documentation / Version Control / DevOps.
Teamwork	Working as a team: Project Management / Code Review Expression and Communicate with abstract concepts / Presentation.

Course Design and Practice

One Week

Git and Gitlab
Engineering Processes
Python / Django
HTML / CSS / JS / Vue

Four Weeks

Survey on Students	Online Preview	Prerequisites	Project based training with agile approach / Defects finding between groups / Discussion and demonstration

One Week

Coding Style
Web FE Fundamental

One Week

Projects Selection
Group Formation
Requirement Analysis
Agile Software Development

DevOps
UI and UX
Testing and Quality
Documentations

Dependency Management
Engineering Security
Survey on OS Projects

Teaching of prerequisite techniques.

Content (6 Days, 4~5 hours / day)				
Web Frontend	HTML、CSS	JavaScript	DOM & Web API	Vue / MVVM
Web Backend	Python	django		
Management	Environment Unification	Dependency Management	Engineering Process	
Git	Git	Git Teamwork		

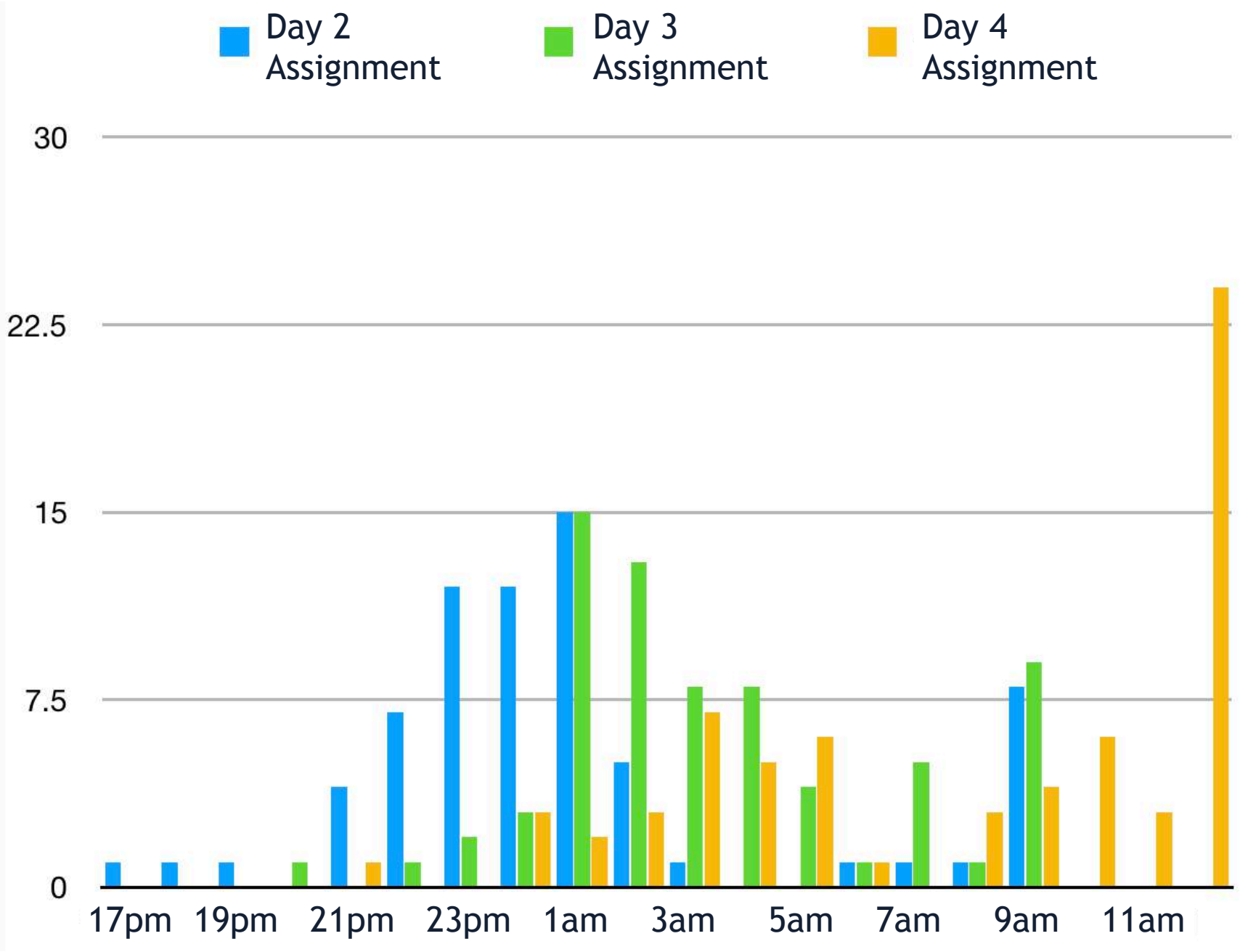
As a supplement to daytime lectures, considering the differences in students' ability, office hours are hosted by engineers to solve questions from students after lectures until 10 p.m.





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Day 2 Assignment
Day 3 Assignment
Day 4 Assignment
30
22.5
15
7.5
0
17pm
19pm
21pm
23pm
1am
3am
5am
7am
9am
11am



Project based training.

2017	Programming Education Game	Instructional Game / Game Editor
	Customer Messaging Platform	Service Desk / Chat bot / Developer Center
	Live Education Platform	Staff Management / Course List / Live Room
2018	Teaching Administration System	Registration / Course Management / Data Recording and Analysis
	English Learning Application	E-books / Community / Content Management
	Science Education Platform	Live Audio Books / Content Management

Group Formation.

- Gender Balanced (at least 1 boy and 1 girl).
- Performance Balanced (excellent in prerequisite techniques).
- Students are asked to score the projects from 1 to 3.
- Decide based on maximizing the mathematical expectation of

$$3N_3 + 2N_2 + N_1$$

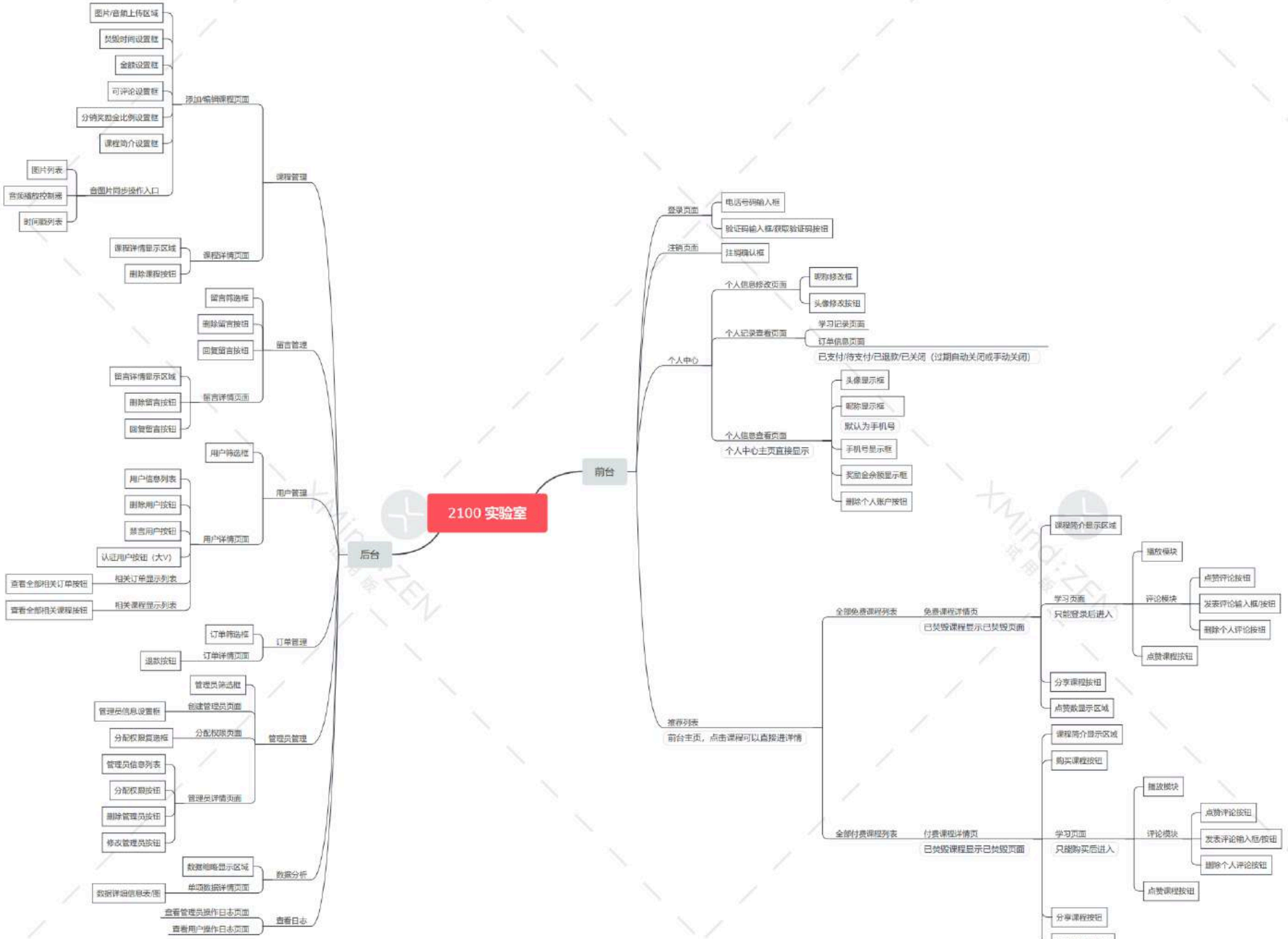
N_i is the number of groups assigned to a project that scored i by the group.

Requirement Analysis.

- Simple Tools: Mind Map / User Stories.
- Face to Face Communication with Real Clients
  - The expressions of real clients are much more equivocal, paradoxical, and misleading.
  - Moderators drop hints to students and encourage them to discover vague statements during the communication.
- Output: user stories, mind maps, and prototypes.






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可评论设置框
分销奖励金比例设置框
课程简介设置框
音图片同步操作入口
图片列表
音频播放控制器
时间戳列表
课程详情页面
课程详情显示区域
删除课程按钮
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用户筛选框
用户详情页面
用户信息列表
删除用户按钮
禁言用户按钮
认证用户按钮（大V）
相关订单显示列表
查看全部相关订单按钮
相关课程显示列表
查看全部相关课程按钮
订单管理
订单筛选框
订单详情页面
退款按钮
管理员管理
管理员筛选框
创建管理员页面
管理员信息设置框
分配权限页面
分配权限复选框
管理员详情页面
管理员信息列表
分配权限按钮
删除管理员按钮
修改管理员按钮
数据分析
数据概略显示区域
单项数据详情页面
数据详细信息表/图
查看日志
查看管理员操作日志页面
查看用户操作日志页面
前台
登录页面
电话号码输入框
验证码输入框/获取验证码按钮
注销页面
注销确认框
个人中心
个人信息修改页面
昵称修改框
头像修改按钮
个人记录查看页面
学习记录页面
订单信息页面
已支付/待支付/已退款/已关闭（过期自动关闭或手动关闭）
个人信息查看页面
个人中心主页直接显示
头像显示框
昵称显示框
默认为手机号
手机号显示框
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发表评论输入框/按钮
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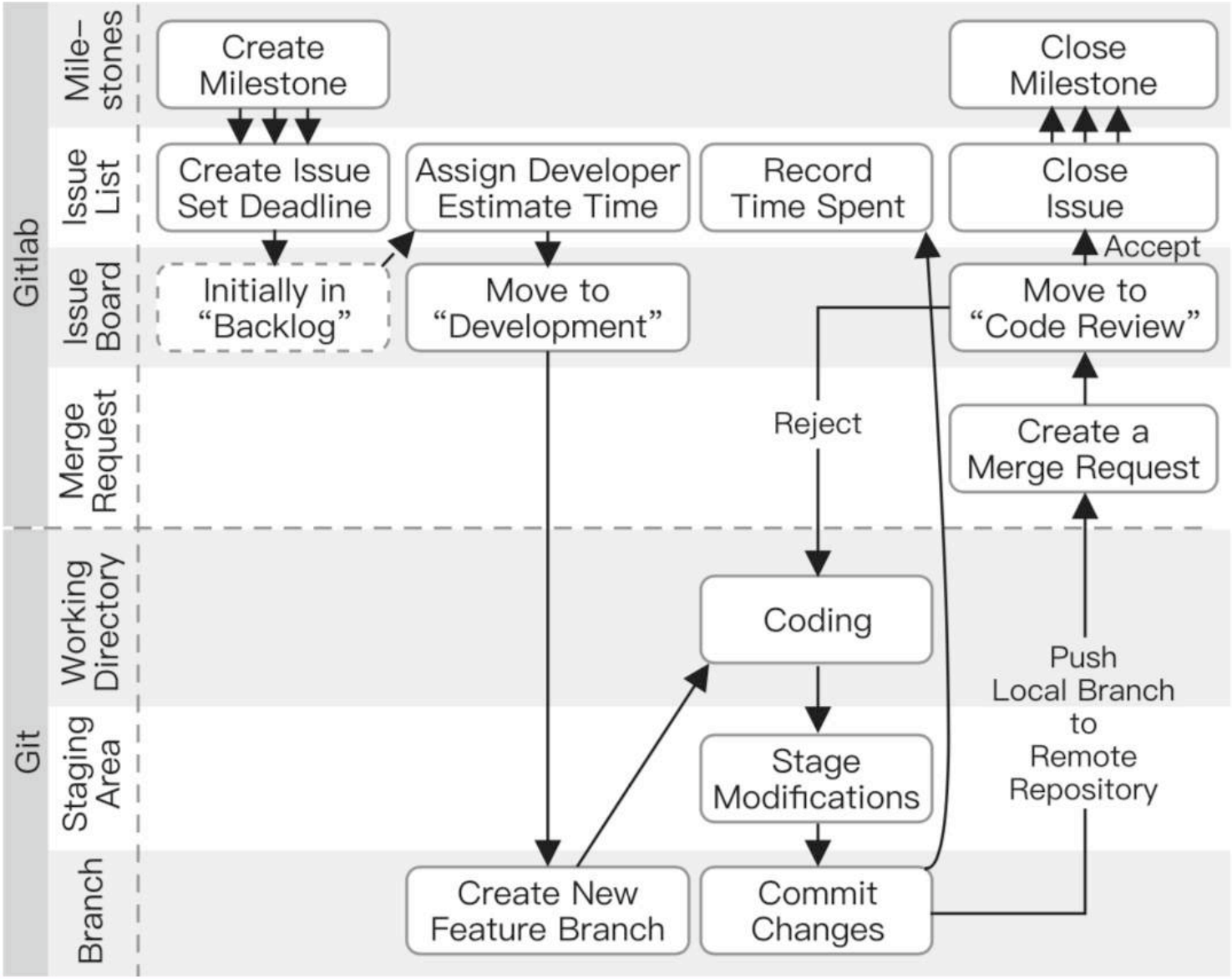
Open Source Understanding and Third-party Dependencies.

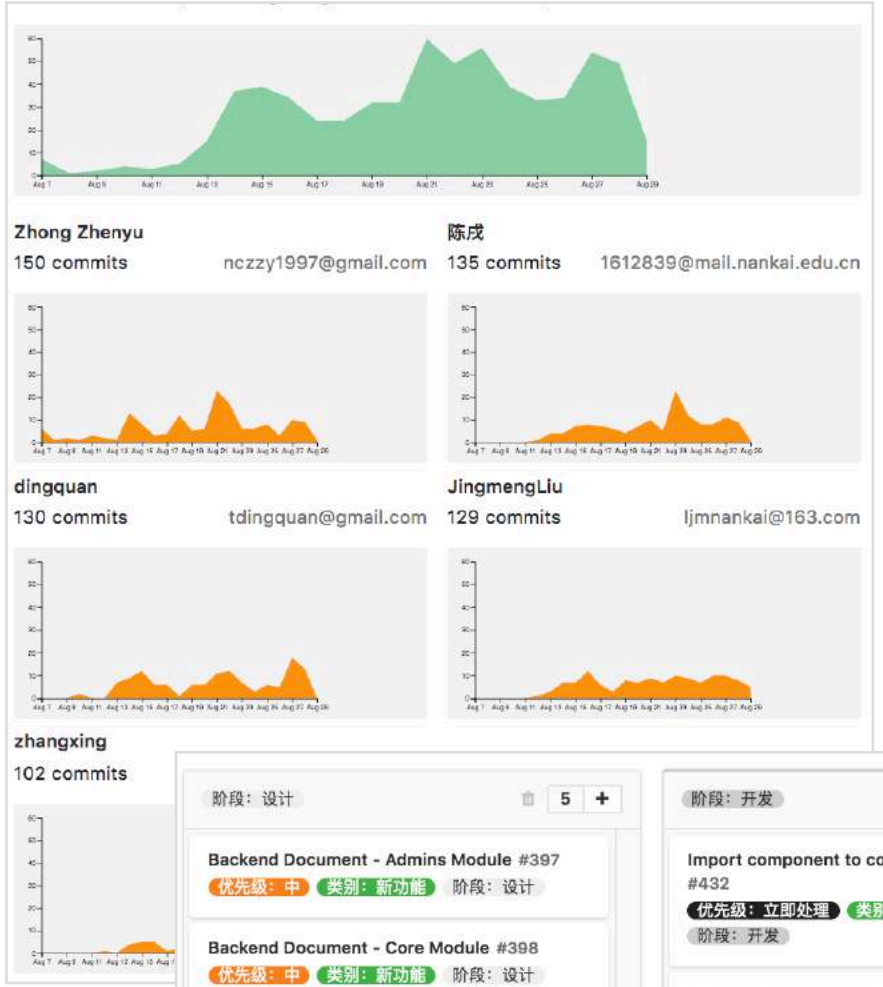
- Find required functions.
- Survey and trade-off on possible solutions:
  - “Implement features from scratch”.
  - “third-party open source projects as dependencies”.
- Emphasis on license issue of softwares.
- Correct practice of dependency management.

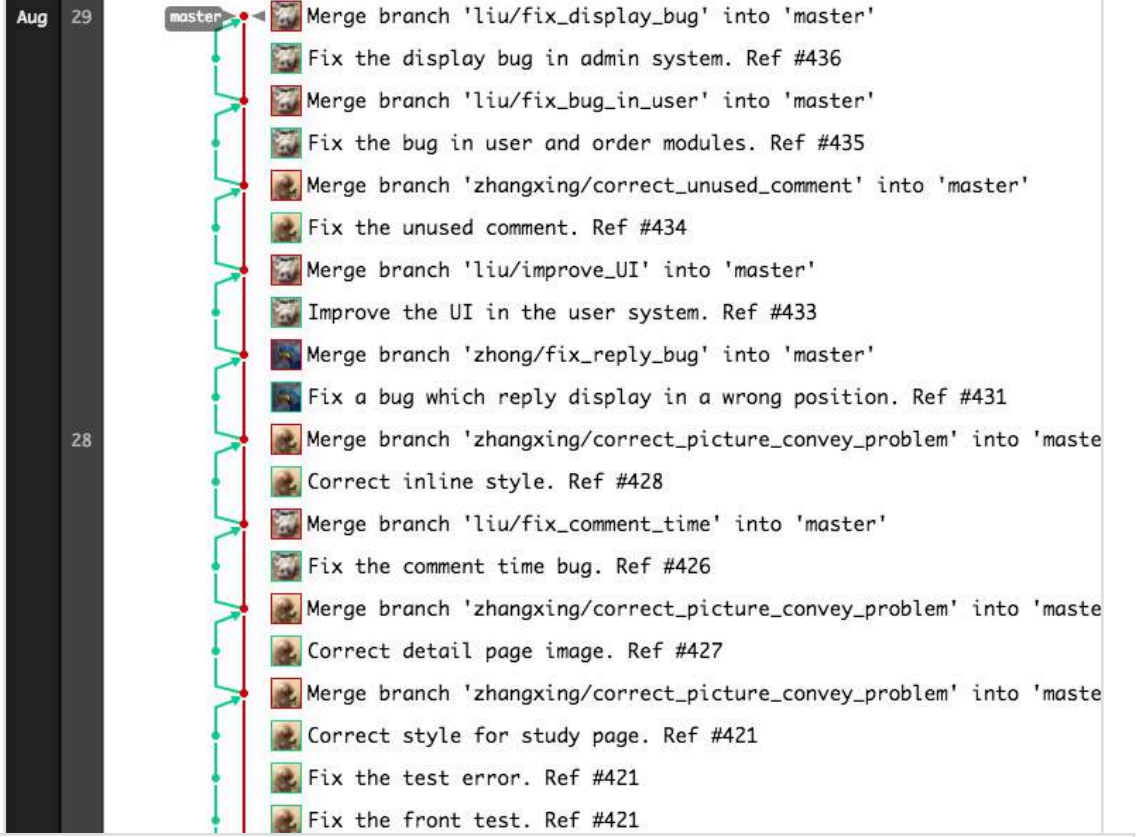
Agile Software Development.

- Git.
- Gitlab.

- Time Tracking.
- Coaching Bot.
- Internal Evaluation.







阶段：设计 5

- Backend Document - Admins Module #397 — 优先级：中 · 类别：新功能 · 阶段：设计
- Backend Document - Core Module #398 — 优先级：中 · 类别：新功能 · 阶段：设计
- Backend Document - Courses Module #399 — 优先级：中 · 类别：新功能 · 阶段：设计
- Backend Document - Customers Module #400 — 优先级：中 · 类别：新功能 · 阶段：设计
- Backend Document - Data Module #401 — 优先级：中 · 类别：新功能 · 阶段：设计

Showing all issues

阶段：开发 3

- Import component to coursedetail page #432 — 优先级：立即处理 · 类别：新功能 · 阶段：开发
- Correct the admin management UI style #437 — 优先级：立即处理 · 类别：改进 · 阶段：开发
- Improve the UI in course detail page #438 — 优先级：高 · 类别：改进 · 阶段：开发

Showing all issues

阶段：评审与测试 0

Showing all issues

Closed 432

- Set up basic frontend and backend structure for the project. #1 — 优先级：高 · 类别：新功能 · 阶段：评审与测试
- Make all the lint files. #2 — 优先级：高 · 类别：新功能 · 阶段：评审与测试
- A problem related to corsheaders needs to be fixed. #4 — 优先级：立即处理 · 类别：BUG · 阶段：设计
- Make git hook files. #3 — 优先级：高 · 类别：新功能 · 阶段：评审与测试
- The setup script is unable to be executed

JIT



Testing and Software Quality.

- Unit test.
- Lint: Style checking.
- Git hook: pre-commit and pre-push trigger.
- CI / CD pipeline.
- Documentation.

passed **Job #5376** in pipeline **#2418** for 15c8

```
LOG LOG: 'thisis test:    123'
    ✓ 123
LOG LOG: 'thisis test:    null'
    ✓ null
LOG LOG: 'object'

  Obj2JSON()
    ✓ {"a":"Hello","b":"World"}
LOG LOG: 'string'
    ✓ Hello World
LOG LOG: 'object'
    ✓ null

  getFormatTime()
    ✓ 不传入参数
    ✓ 传入0
    ✓ 秒数小于10
    ✓ 秒数不小于10
    ✓ 分钟数小于10
    ✓ 分钟数不小于10
    ✓ 小时数小于10
    ✓ 小时数不小于10

PhantomJS 2.1.1 (Linux 0.0.0): Executed 112 of 112 S
TOTAL: 112 SUCCESS
```

Pipeline	Commit	Stages
#2591 by latest	master bc69a9ff Merge branch 'romic/backend_docu...	
#2590 by	romic/backend… 996ef58f Done with documents. Ref #307	
#2584 by	master fe80166c Merge branch 'xiaochen/code_style'...	
#2583 by	xiaochen/code… e463b8a5 Modify code style. Ref #306	
#2582 by	master f75c62e8 Merge branch 'romic/fix_stupid_bug...	

09% Statements 393/1686 16.36% Branches 123/752 16.68% Functions 61/368 13.44% Lines 173/1287 16 branches ignored

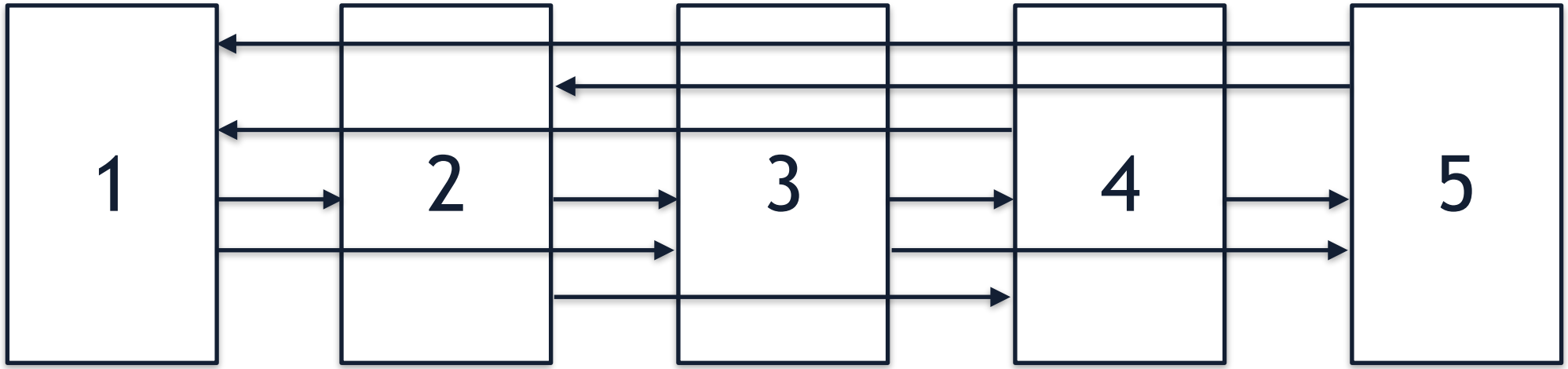
File	Statements		Branches		Functions	
node_modules/vue-schart/src/		31.58%	6/19	100%	4/4	9.0
src/components/		17.84%	207/1160	9.94%	51/513	18
src/module/Enterprise/		90.48%	19/21	100%	8/8	5
src/module/User/		21.3%	23/108	36.36%	8/22	8
src/module/customer_active/		32.5%	13/40	22.22%	4/18	12
src/module/customer_login/		27.36%	23/84	20.51%	8/39	12
src/module/customer_manage/		16.04%	34/212	11.76%	8/68	7.3
src/module/customeractive/		32.5%	13/40	22.22%	4/18	12
src/module/enterprise_active/		44.19%	19/43	42.66%	12/28	27.7
src/module/enterprise_manage/		49.15%	29/59	33.33%	8/24	15.3
src/module/reset_password/		65%	13/20	60%	8/10	33.3



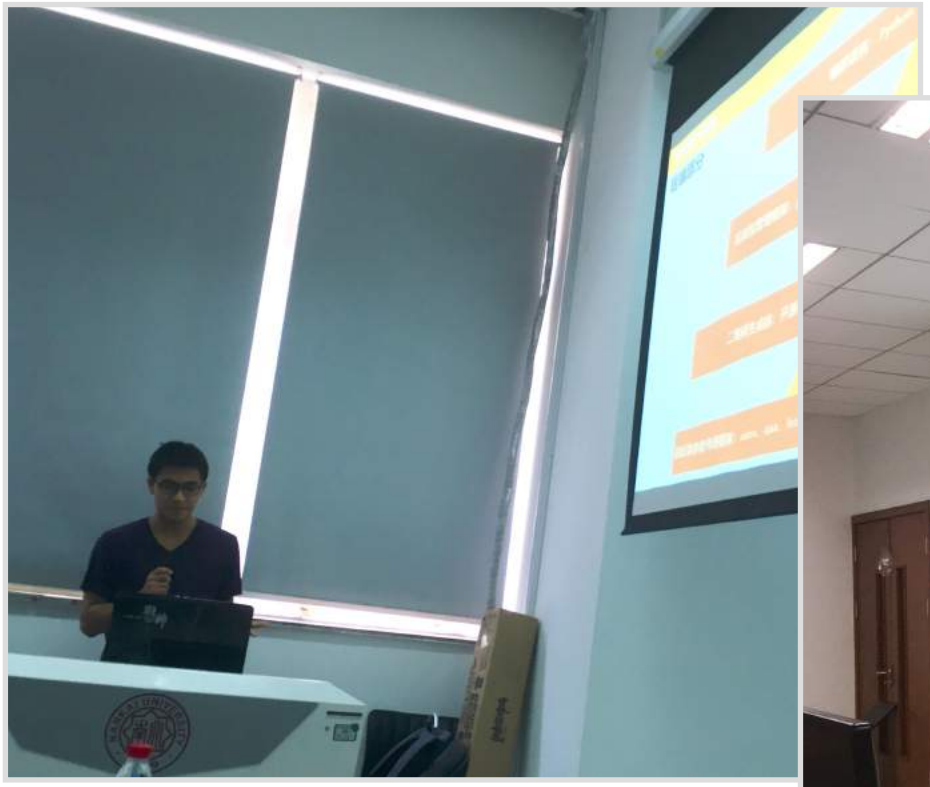


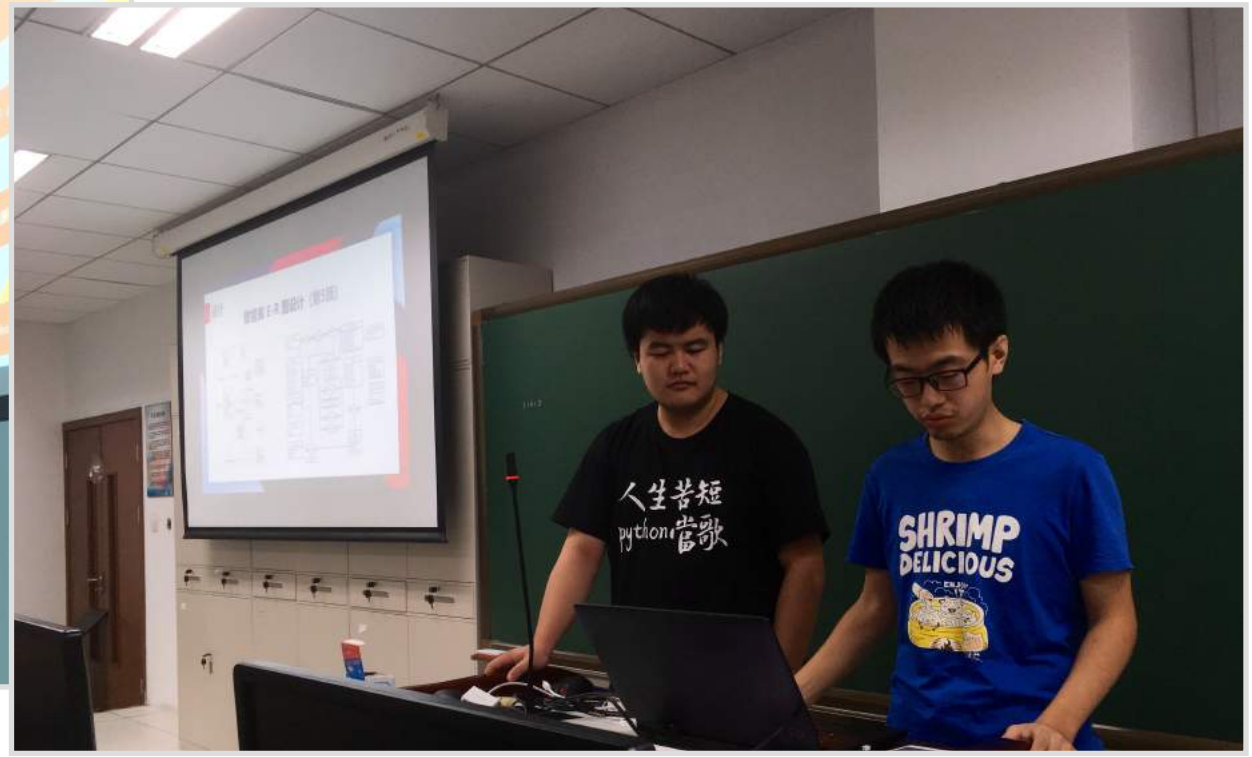
Defects Finding Between Groups.

Each repository is opened to two other groups who have the same client. The students in the two groups are required to act as reporters and check the projects specified to them.



Discussion and Demonstration



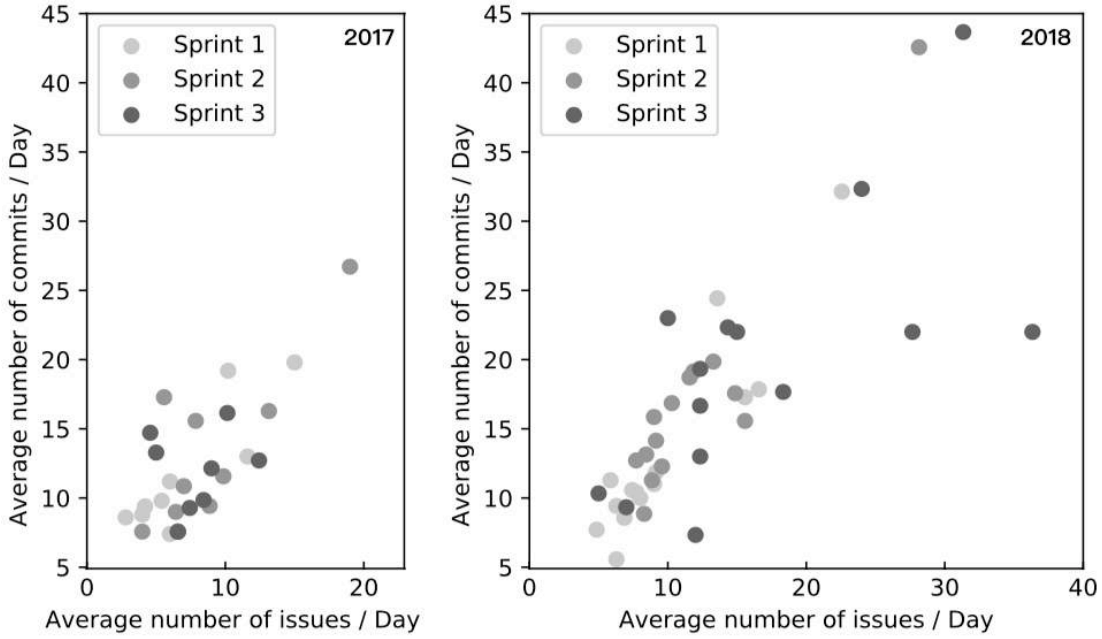


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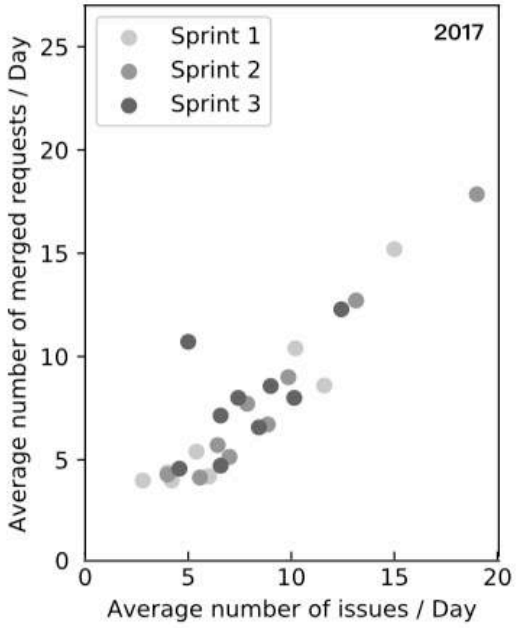
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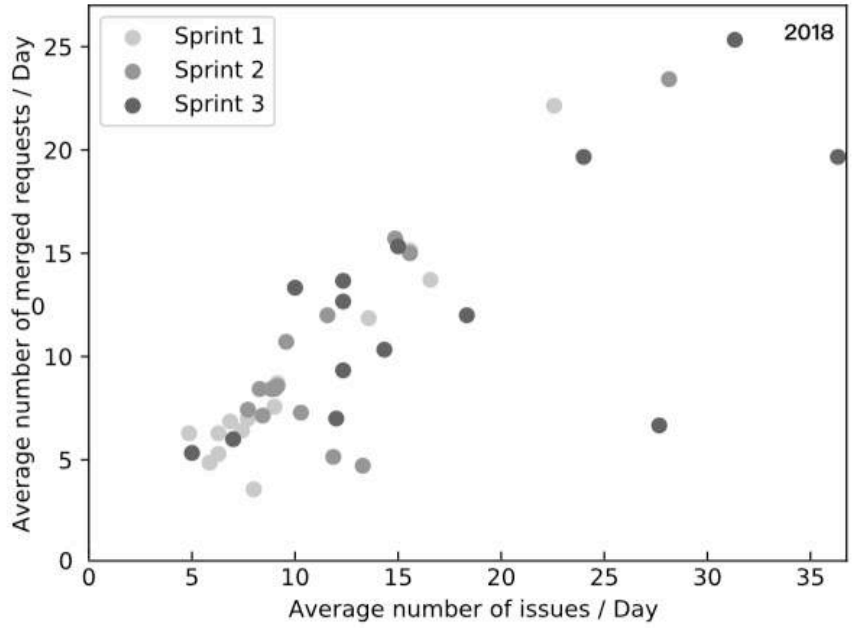
Evaluation and Result

Issues, Merges, Commits



Relation between Issues and Commits

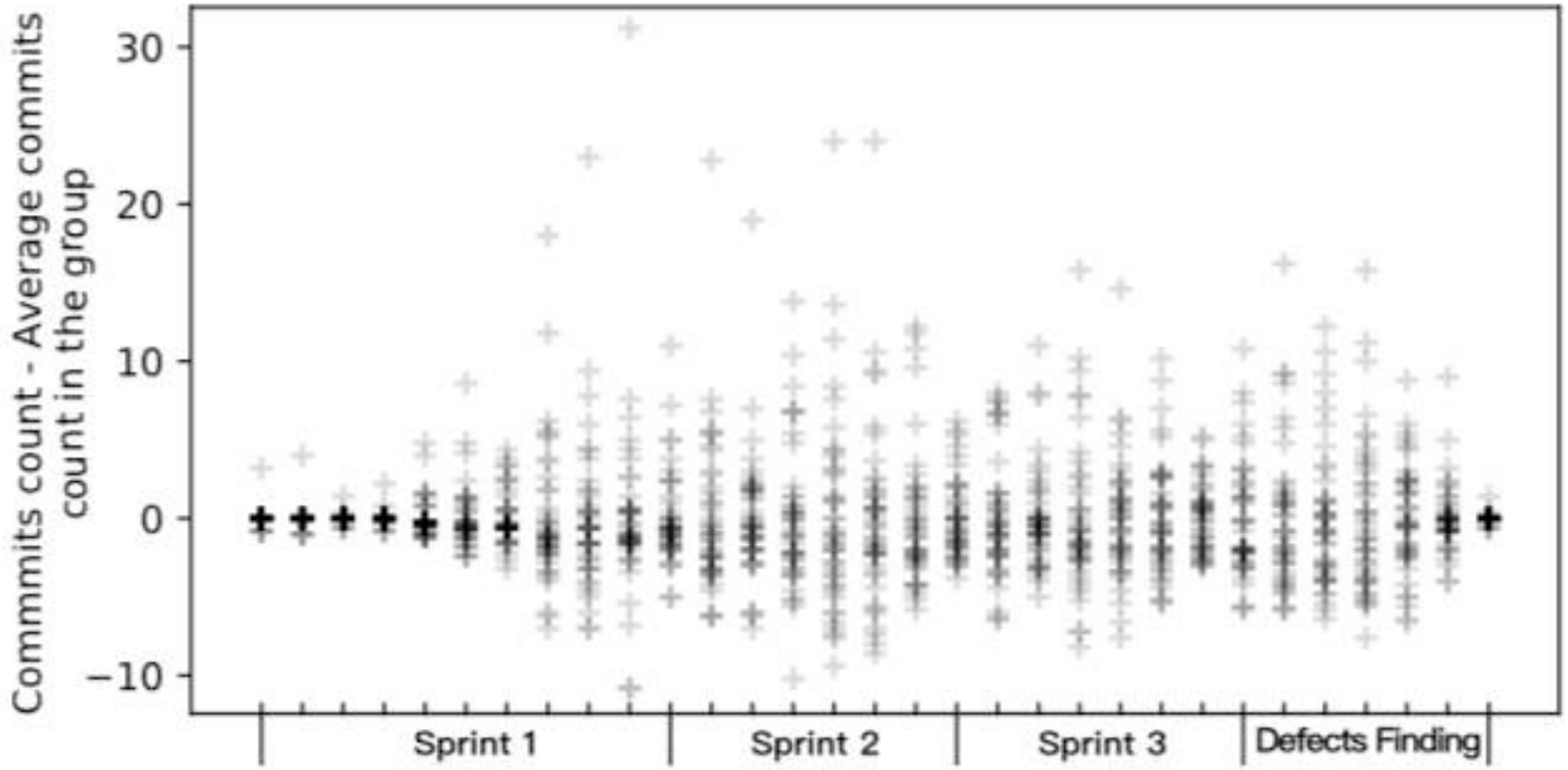




Relation between Issues and Merges to Master Branch

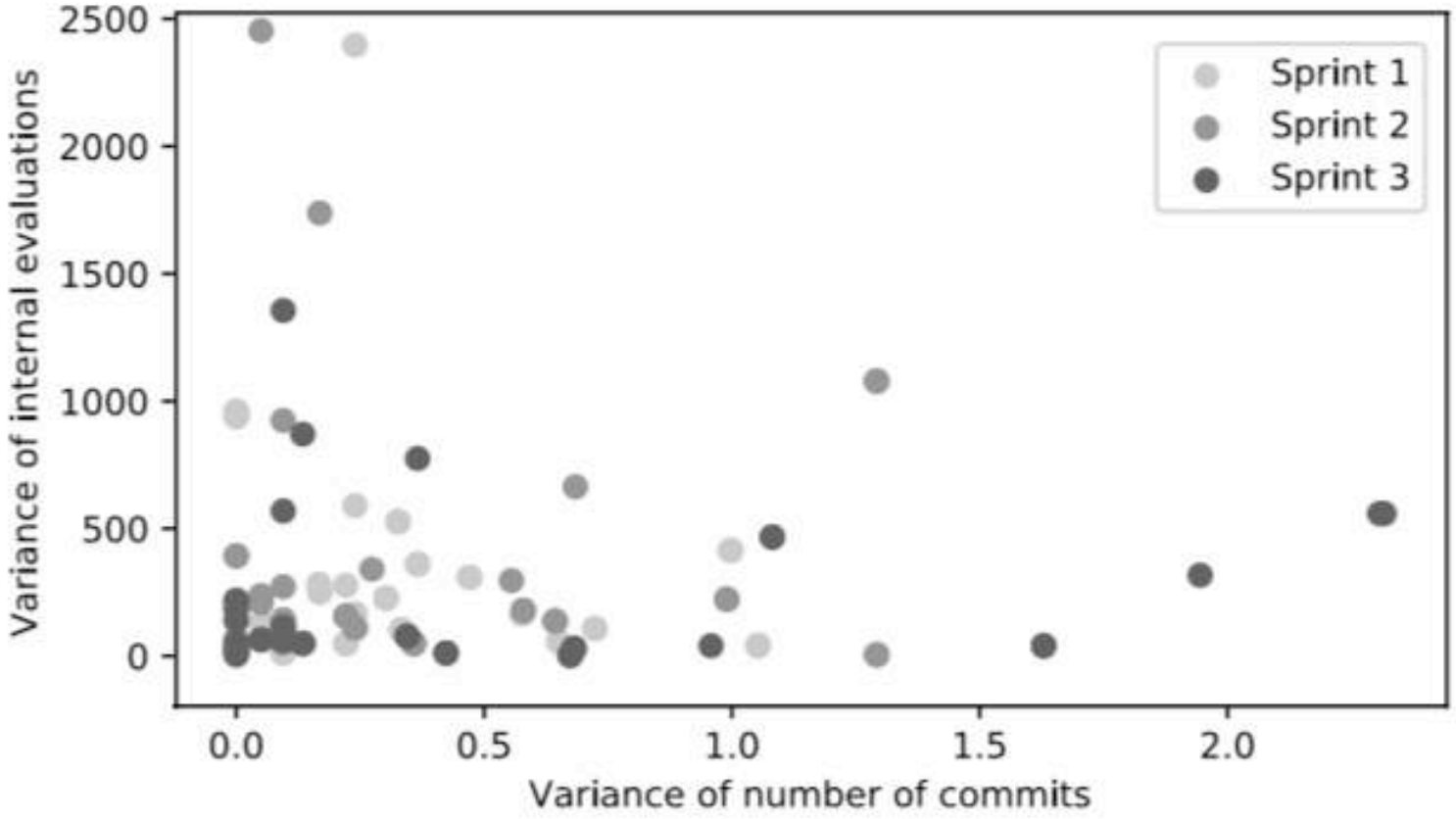
Evaluation and Result

Teamwork - by contribution difference



Evaluation and Result

Teamwork - internal evaluation & commits



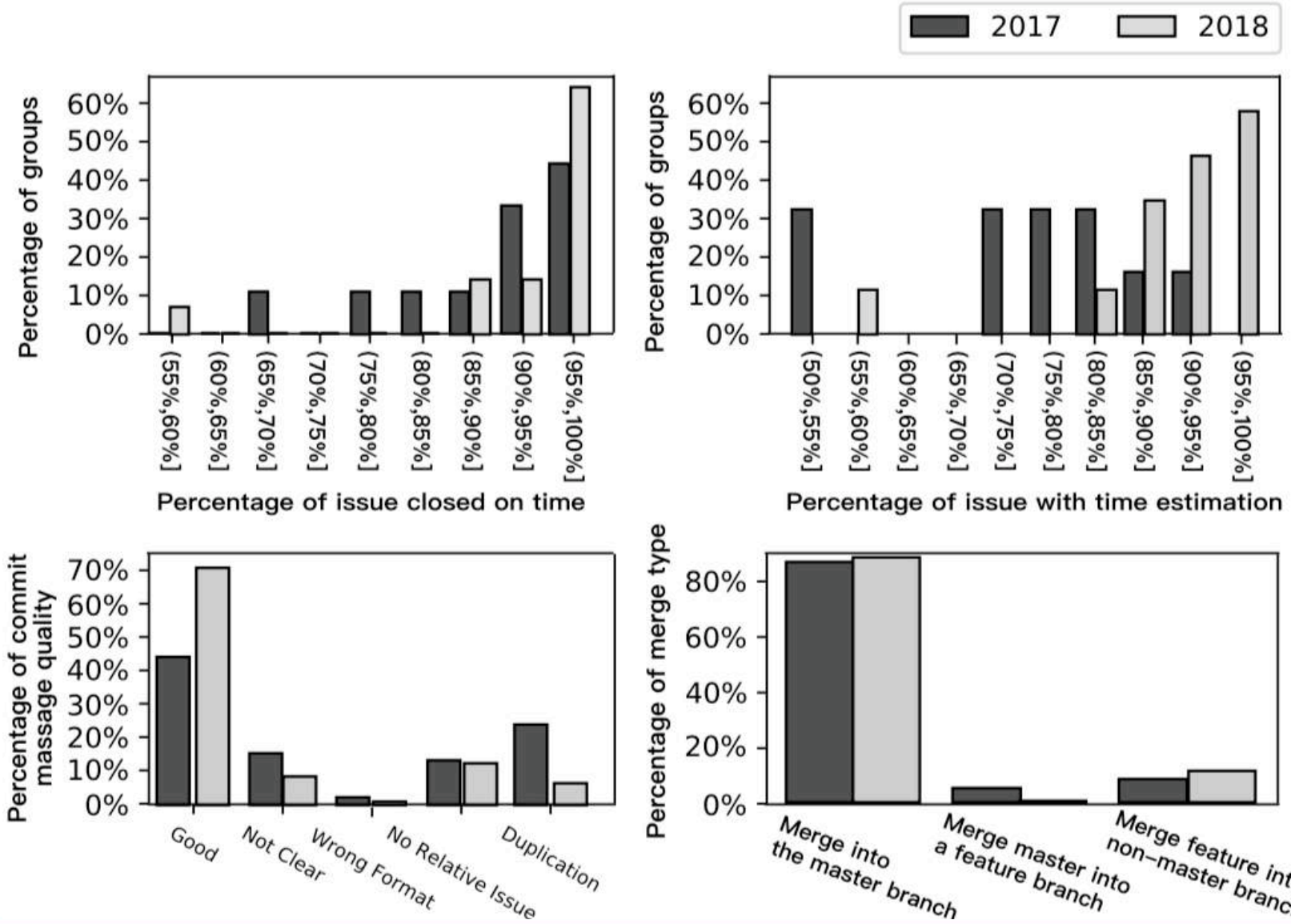
Evaluation and Result

Improvement from 2017 to 2018

- Time estimation of each issue was required to be checked by at least two people before development.
- A “commit-msg” Git hook was recommended to students to check the eligibility of the commit message of each commit.
- “Merge commit with semi-linear history” on the Gitlab to eliminate the possibility of merge errors being recorded in the log of the master branch.

Evaluation and Result

Outcome from 2017 to 2018



Evaluation and Result

Questionnaires

Students are required to score following statements by rating a score from 1 to 5.

1. I am familiar with the code version management tools now.
2. I agree more on the importance of software requirements analysis now.
3. I believe more that software development process is helpful to the effectiveness of development now.
4. I realize the importance of teamwork throughout the course.
5. After training, I am more aware of the importance of coding style and software design.
6. I believe more on the significance of testing after training.
7. I tend to write more documentation in software development after training.
8. Weekly presentations allow us learn new knowledge from others.
9. Cross-group defects finding make me aware of our shortcomings and where we can make further improvements.
10. I will tend to survey and trade off between reuse third-party dependencies and coding from scratch in the future.

Evaluation and Result

Questionnaires outcome

Year	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
2017	4.48	4.60	4.79	4.43	4.81	4.52	4.40	4.62	4.24	4.52
2018	4.53	4.86	4.80	4.50	4.80	4.46	4.49	4.57	4.47	4.53

- Highlights: be convinced of on the importance of requirement analysis, development process, coding style and software design.
- Future Works: awareness on software testing and software documentation and teamwork.



Thank You!
Q & A

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